

REVIEW

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Efficacy and safety of sulforaphane in schizophrenia: a systematic review and meta-analysis of randomized controlled trials

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Abstract

Introduction Sulforaphane, an isothiocyanate derived from cruciferous vegetables (e.g., broccoli sprouts), has been explored for its antioxidant and anti-inflammatory properties. This is the first systematic review and meta-analysis to explore the therapeutic potentials of sulforaphane in schizophrenia.

Methods We searched PubMed, Scopus, Web of Science, and Cochrane Central for studies from inception to April 2025. We included randomized controlled trials (RCTs) evaluating the efficacy of sulforaphane in schizophrenia. The primary outcomes were changes in the Positive and Negative Syndrome Scale (PANSS) total and its subscales. Secondary outcomes included cognitive measures, metabolic markers, and safety.

Results Four RCTs with 369 schizophrenia patients were included. Sulforaphane did not significantly improve PANSS total or positive symptom scores at the latest follow-up (ranging from 24 weeks to 18 weeks) or at a consistent 12-week time point. However, a modest improvement in negative symptoms was found at 12 week- time point (MD= -1.06; 95% CI: -1.95 to -0.16; $p=0.02$), which was not maintained at the latest follow-up. General psychopathology scores improved significantly (MD= -1.5; 95% CI: -2.78 to -0.23; $p=0.02$). No cognitive benefits were observed. Sulforaphane led to significant reductions in metabolic markers, including LDL, triglycerides, and cholesterol. Discontinuation rates were lower in the sulforaphane group (RR= 0.68; 95% CI: 0.49 to 0.95; $p=0.02$).

Conclusion The study provides initial insights into sulforaphane's potential therapeutic effects in schizophrenia, showing modest improvements in general psychopathology and negative symptoms, with favorable metabolic changes and lower discontinuation rates. Due to limited data and heterogeneity, the study findings should be interpreted with caution.

Clinical trial number Not applicable.

Keywords Meta-analysis, Metabolism, PANSS, Psychopathology, Schizophrenia, Sulforaphane

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Introduction

Schizophrenia is one of the most prevalent long-term disabling psychiatric disorders, affecting around 0.7% to 1% of the general population [1]. It involves two main psychiatric manifestations: positive and negative symptoms. The positive symptoms manifest as hallucinations, delusions, and disorganized speech or behavior, while the negative symptoms consist of cognitive impairment with reduced motivation and expressiveness factors that affect memory and mental processing [2, 3]. In addition to these symptoms, schizophrenia involves broader symptom domains such as cognitive deficits and depressive symptoms [4, 5]. Schizophrenia roughly disrupts daily life, with around 24–50% of individuals experiencing suicidal ideation and about 20–40% making suicide attempts [6]. Patients with schizophrenia have a mortality rate of 2–3-fold compared to healthy people, with a reduced expected life of 10–20 years [7].

The main treatment guidelines emphasize the use of first-generation antipsychotics, such as haloperidol and chlorpromazine, as well as second-generation agents like quetiapine and olanzapine [8]. There are also psychosocial interventions, such as psychoeducation therapy, family interventions, and cognitive-behavioral therapy, which act as add-on treatments due to their beneficial effects [2, 8]. The main goal of antipsychotics is restoring the patient's normal daily functioning, followed by maintenance doses of therapy to prevent the risk of hospitalization and improve quality of life [2]. However, current pharmacotherapies have shown a modest effect in handling cognitive impairment and negative symptoms, urging further investigation of adjunctive agents [9, 10]. Also, approximately 30% of patients undergo treatment resistance in addition to the different side effects, such as extrapyramidal symptoms, sedation, and various metabolic changes [11–13].

Sulforaphane, a compound derived from cruciferous vegetables, has recently been proposed as a potential therapeutic agent owing to its antioxidant and anti-inflammatory properties [14]. The primary mechanism involves the activation of nuclear factor erythroid 2-related factor 2 (Nrf2), which regulates the cellular antioxidant response by enhancing glutathione expression, an essential antioxidant whose depletion is implicated in the pathophysiology of schizophrenia [15, 16]. Additionally, it inhibits neuroinflammatory pathways by mitigating the release of pro-inflammatory cytokines [17]. According to a preclinical study by Shirai et al. 2012, sulforaphane has been proposed to have antipsychotic activity in an animal model of schizophrenia due to the multidimensional mechanisms of action [18].

A single-arm trial by Shiina (2015) involving 10 patients with schizophrenia found that sulforaphane did not improve positive or negative symptoms but showed

potential to improve cognitive function [19]. Several randomized controlled trials (RCTs) estimated the efficacy of sulforaphane for patients with schizophrenia with conflicting results [20–23]. Dickerson (2021) found no significant effects of sulforaphane on any clinical outcomes [23]. Hei (2022) also found no significant improvement in overall symptoms but reported improvements in specific cognitive domains, including working memory, reasoning, and verbal learning [22]. Both Huang (2025) and Chen (2025) found that sulforaphane significantly reduced PANSS negative symptom scores but with no significant effects on cognitive outcomes [20, 21]. A recent meta-analysis in autism by Guo et al. showed that sulforaphane significantly improved irritability and hyperactivity symptoms in autistic patients [24]. Several nutraceuticals and phytochemicals have been explored as adjunctive treatments for schizophrenia. A recent network meta-analysis synthesized evidence from 22 interventions and reported benefits for some, such as citicoline, N-acetylcysteine, and sarcosine [25]. Sulforaphane was included in this network but represented by only one RCT, limiting confidence in its estimated efficacy. So, to resolve this discord, we conducted this systematic review and meta-analysis of randomized controlled trials RCTs to assess the efficacy of sulforaphane as an adjunctive therapy for patients with schizophrenia. To the best of the author's knowledge, no previous systematic review and meta-analysis provide precise evidence regarding its therapeutic potential and safety profile that could potentially improve the overall treatment outcome better than the traditional antipsychotics.

Methods

Protocol

We followed the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA statement) guidelines when reporting this study [26]. This manuscript was conducted in adherence to the Cochrane Handbook of Systematic Reviews of Interventions [27]. All authors worked independently at every stage of the study. This study was registered on PROSPERO (CRD42024603653).

Search strategy and data source

We conducted a comprehensive search in April 2025 across the following databases: PubMed, Web of Science, Scopus, and the Cochrane Library for eligible studies. The search approach used MeSH (medical subject headings) and their synonyms. The search terms used were “Sulforaphane” AND “Schizophrenia”. The search terms were combined using Boolean operators to ensure comprehensive retrieval of relevant studies. Additionally, Google Scholar results and references of included studies were manually screened for any possible studies to

include. The detailed search terms used for each database are illustrated in Supplementary Table 1.

Eligibility criteria

Studies were selected based on predefined inclusion and exclusion criteria. Studies satisfying the following criteria were included: [1] Population: patients diagnosed with schizophrenia according to validated diagnostic tools (e.g., DSM or ICD criteria) [2], Intervention: sulforaphane as an adjunct to antipsychotics [3], Comparator: placebo adjunctive to antipsychotics or alone [4], Outcomes: any efficacy or safety outcomes related to schizophrenia [5], Study design: randomized controlled trials (RCTs). Studies were excluded if they were: [1] case reports, case series, observational studies, and conference abstracts [2], animal studies or in vitro studies [3], reviews, book chapters, theses, letters, and overlapped datasets, and [4] non-English studies.

Screening and study selection process

Two independent authors (NF and YK) used Rayyan for semi-automated screening of the literature search results [28]. Studies were screened in two phases; we screened the titles/abstracts for potential clinical studies in the first phase. In the second phase, we retrieved the full-text articles of the selected abstracts for further eligibility screening. Any disagreements between the two reviewers were resolved through consultation with the first author (OK).

Data extraction

Data extraction was independently performed by two authors (AS and OY) using a standardized data extraction form developed in Microsoft Excel. Extracted data included study characteristics, such as author(s), year of publication, country and locations, study design, sample size, inclusion criteria, treatment protocol, comparator, outcome measures, and key findings of included studies. The Second sheet included baseline characteristics of the included studies such as, age, sex, and duration of illness. A third outcome sheet was specifically designed to collect the outcomes of each study. Any disagreements in data extraction were resolved by consensus or consultation with the first author (OK).

Risk of bias assessment

The quality of the studies included was evaluated by two independent authors (AS and MM) using the Cochrane risk of bias assessment tool for RCTs version two (ROB2) [29]. Bias potential was evaluated across five domains, including (1) bias due to randomization process (2) bias due to deviations from intended interventions (3) Bias due to missing outcome data (4) Bias in measurement of the outcome (5) Bias in selection of the reported result.

The overall judgment for each domain was classified into one of three levels: low, moderate, or high risk of bias. Conflict between authors was resolved through mutual agreement or by consulting the primary investigator (OK).

Outcome definition

The primary efficacy outcomes included changes in the total Positive and Negative Syndrome Scale (PANSS) score and its subscales: PANSS Positive, PANSS Negative, General Psychopathology, PANSS Depression, and PANSS Cognition. PANSS is a 30-item clinician-tool used to assess positive, negative, and general psychopathology symptoms in schizophrenia, and it is the gold standard for measuring treatment response [30]. Secondary outcomes were the effects of sulforaphane on cognitive assessments, the metabolic markers, discontinuation rates, and adverse events. Cognitive function was assessed using the MATRICS Consensus Cognitive Battery (MCCB), assessing total score and specific domains (e.g., verbal learning, spatial working memory, reasoning/problem solving) [31]. Metabolic markers, including low-density lipoprotein (LDL), high-density lipoprotein (HDL), total cholesterol (TC), triglycerides (TG), and fasting glucose (FG), were measured using standard laboratory assays in fasting blood samples.

Data synthesis and statistical analysis

We analyzed data using RevMan software, version 5.4. For continuous data, the mean difference (MD) between the two groups, along with its standard deviation (SD) and total number of patients, were pooled using the inverse variance method with a random effect model. For dichotomous data, the frequency of events and the total number of patients were pooled as risk ratios (RR) between the two groups, also using the inverse variance method with a random effect model. Heterogeneity was assessed using the chi-square test and I^2 statistic; a p-value from the chi-square test < 0.1 and $I^2 > 50\%$ was considered indicative of significant heterogeneity. We conducted the analysis based on the primary endpoint of each included studies and a consistent time across all studies. Subgroup analyses were conducted according to the reported daily sulforaphane dose (~ 66 $\mu\text{mol/day}$, ~ 100 $\mu\text{mol/day}$, and 136 $\mu\text{mol/day}$). Sensitivity analysis was conducted by excluding one study in each scenario to explore sources of heterogeneity.

Publication bias

In accordance with Egger et al., we did not assess publication bias using Egger's test for funnel plot asymmetry. The Egger test requires a minimum of 10 included studies to be applicable [32].

Results

Study selection

A systematic search of relevant databases identified a total of 230 articles. After removing duplicates, we excluded 152 studies during the initial screening of titles and abstracts as they did not fulfill the pre-defined inclusion criteria. This process resulted in 22 studies being selected for full-text review. Upon further evaluation, 17 studies were excluded for specific reasons, which are detailed in the PRISMA flow diagram Fig. 1. Consequently, 4 studies met all eligibility criteria and were included in the final meta-analysis.

Summary and baseline characteristics

Four RCTs comprising 369 schizophrenia patients were included in the study. Three studies were conducted in China [20–22] and one study in the USA [23]. The

treatment duration varied across studies: two studies lasted 24 weeks, one 22 weeks, and one 18 weeks. The mean age of participants ranged from 23 years to 45 years. Sulforaphane and placebo were adjunct to antipsychotics in all included studies. The doses of sulforaphane administered across the studies ranged from 66 μmol to 136 μmol . Baseline characteristics and a summary of the studies included are shown in Tables 1 and 2.

Risk of bias assessment

A summary and graph of the risk of bias in our included studies are shown in (Fig. 2). Results of the risk of bias assessment showed that the quality of the included studies ranged from low risk to some concerns. Three studies yielded an overall low risk of bias. However, Huang et al. demonstrated some concerns in domain 2 (bias due to deviations from intended intervention).

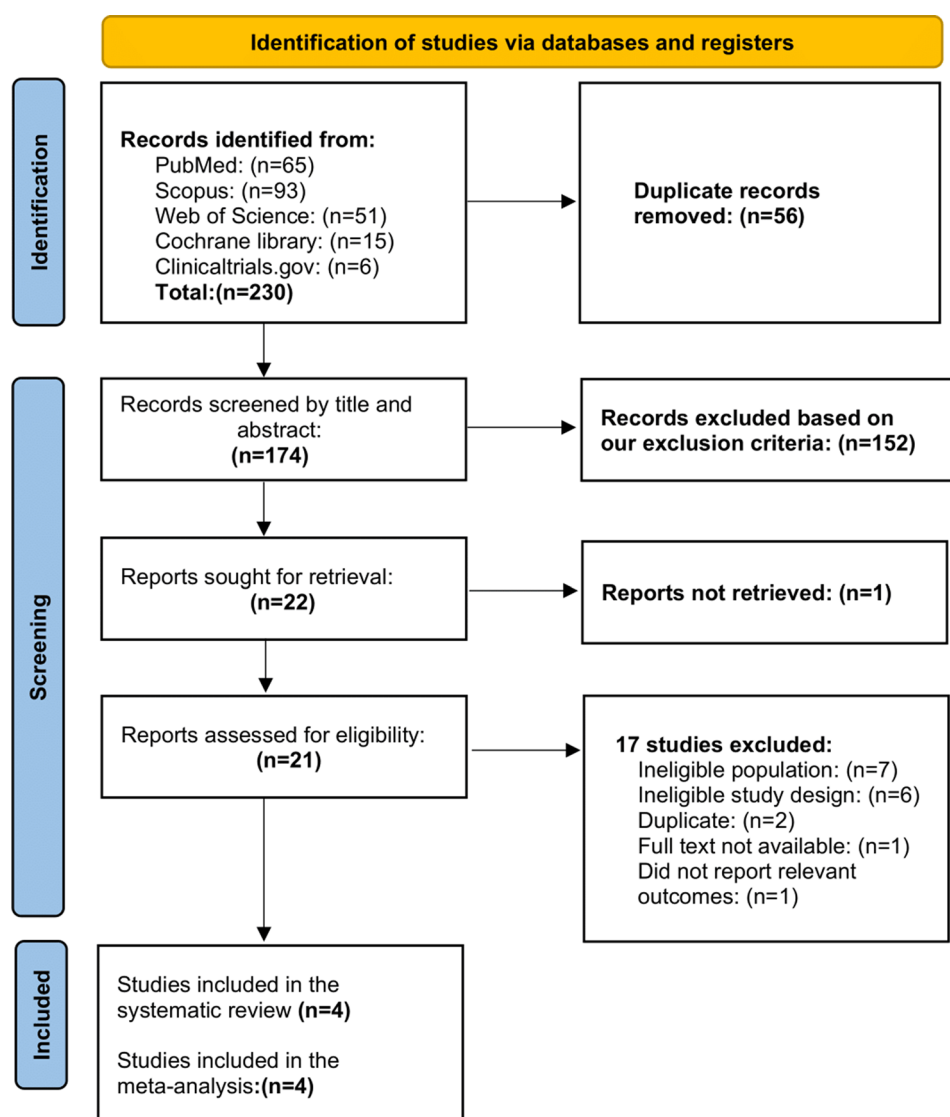


Fig. 1 Prisma flow diagram of the included studies

Table 1 Summary of the included studies

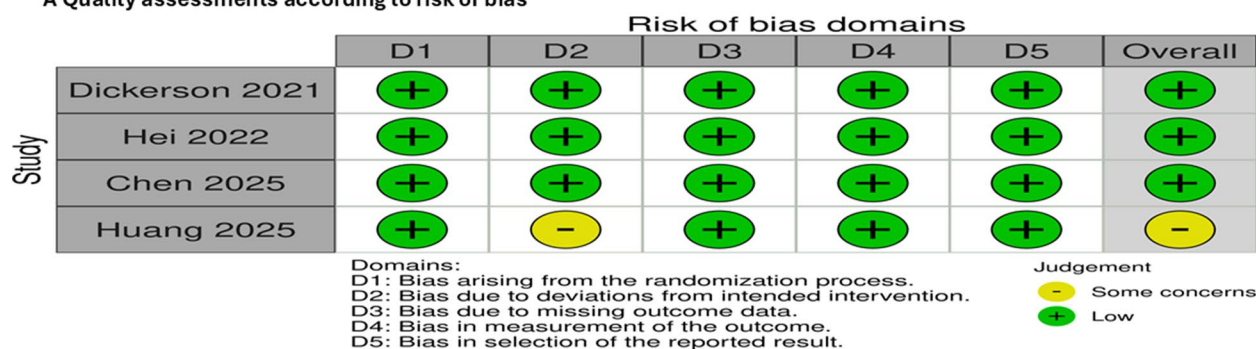
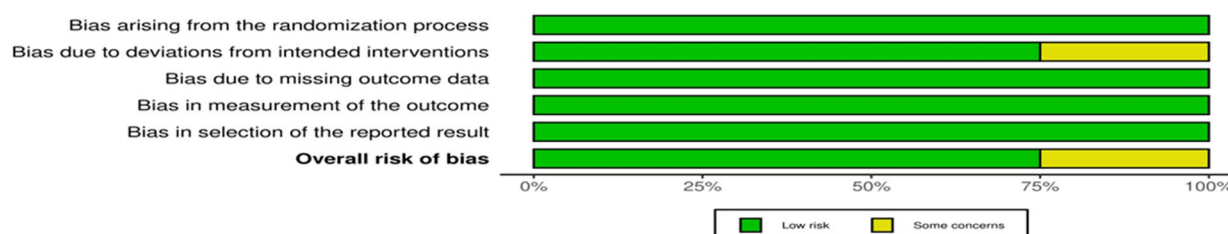
Study ID	Location	Year	Study design	Population	Sam- ple size	Intervention	Dose	Comparator	Treat- ment duration	Outcome measures	Conclusion
Dick- erson 2021	USA	2017– 2019	RCT	Adults aged 18–65 with schizophrenia or schizoaffect- ive disorder, showing moderate psychotic symptoms and stable on antipsychotic medication	64	Sulfora- phane plus antipsychotics	100 μmol	Placebo plus antipsychotics	18 weeks	PANSS and MCCB	Sulforaphane showed no sig- nificant effects on symptoms or cognition in schizophrenia over 18 weeks. It was safe and well-toler- ated, with no major differences in adverse events between groups.
Hei 2022	China	2016– 2019	RCT	Adults aged 18–50 with early- stage schizophrenia and a PANSS score of 75 or higher were included, diagnosed via clinical interviews using DSM-5 criteria.	172	Sulfora- phane plus antipsychotics	66, 99 μmol	Placebo plus antipsychotics	22 weeks	MATRICS Compos- ite score, change in MATRICS Domain scores, PANSS Total Scores and change in side-effects	The study showed improvements in working memory, attention, and processing speed in early episode schizophrenia, despite no overall cognitive enhancement, indicating potential benefits of sulforaphane and the need for further research.
Chen 2025	China	2020– 2022	RCT	Adults aged 18–55 with long- standing, stable schizophrenia (10–30 years) were included. They had no recent medication changes, scored > 3 on at least one PANSS negative symptom item.	43	Sulfora- phane plus antipsychotics	136 μmol	Placebo plus antipsychotics	24 weeks	PANSS Negative Symp- toms, MCCB Cognitive Battery, Treatment Emergent Symptom Scale (TESS), Clinical Global Impression (CGI), and Global Assessment of Functioning (GAF).	Sulforaphane significantly reduced negative symptoms in chronic schizophrenia and was well toler- ated but had limited effects on overall cognition and functioning.
Huang 2025	China	2020– 2022	RCT	Adults aged 18–50 with schizo- phrenia for 10 years or less, showing prominent negative symptoms, on stable treatment with no more than two antipsy- chotics, were included.	90	Sulfora- phane plus antipsychotics	136 μmol	Placebo plus antipsychotics	24 weeks	PANSS Negative Symptoms, PANSS Total Score, PANSS 5-Factor Scores, Clinical Global Impression (CGI) Scale, Treatment Emergent Symptom Scale (TESS).	Sulforaphane added to antipsy- chotics significantly reduced nega- tive symptoms in schizophrenia over 24 weeks, was well tolerated, and showed promise as an effec- tive adjunctive treatment.

Abbreviations: RCT: Randomized controlled trial; PANSS: Positive and Negative Syndrome Scale; MCCB: MATRICS Consensus Cognitive Battery; DSM-5: Diagnostic and Statistical Manual of Mental Disorders, Fifth Edition; TESS: Treatment Emergent Symptom Scale; CGI: Clinical Global Impression; GAF: Global Assessment of Functioning

Table 2 Baseline of population of the included studies

Study ID	Group name	Age (SD)	Male (%)	BMI	Duration of illnesses (months)	PANSS total	PANSS positive subscale	PANSS negative subscale	PANSS General psychopathology
Dickerson 2021	Sulforaphane	42.7 ± 12.3	24 (75%)	34.7 (8.8)	N/A	84.1 ± 13.5	22.4 (± 5.4)	21.3 (± 4.8)	40.5 (± 6.8)
	Placebo	45.8 ± 11.7	25 (78%)	31.8 (± 5.2)	N/A	83 ± 12.3	21.9 (± 5.3)	22.2 (± 4.8)	38.9 (± 6.6)
Hei 2022	Sulforaphane HD	24.64 ± 6.39	20 (38%)	21.71 ± 4.2	14.37 ± 13.51	87.60 ± 14.90	15.32 ± 5.1	21.47 ± 7.1	N/A
	Sulforaphane LD	24.73 ± 7.57	23 (48%)	21.26 ± 3.4	16.66 ± 13.08	88.79 ± 14.28	14.77 ± 5.6	21.15 ± 7.1	N/A
	Placebo	23.18 ± 6.43	24(48%)	21.59 ± 3.5	13.79 ± 11.48	91.69 ± 16.71	15.00 ± 5.04	22.04 ± 7.1	N/A
Chen 2025	Sulforaphane	42.50 (7.96)	21 (75%)	24.17 (4.37)	N/A	65.29 (7.45)	13.11 (3.00)	22.04 (3.37)	30.14 (4.34)
	Placebo	44.36 (7.48)	10 (71%)	23.51 (3.25)	N/A	66.00 (9.45)	13.21 (2.81)	22.86 (6.34)	29.93 (4.12)
Huang 2025	Sulforaphane	23.75 ± 6.21	28 (52%)	N/A	44.2 ± 35.22	73.83 ± 11.88	12.06 ± 3.83	24.70 ± 4.24	N/A
	Placebo	23.68 ± 6.32	12(50%)	N/A	43.43 ± 47.64	76.67 ± 12.60	16.33 ± 5.95	24.42 ± 4.03	N/A

All qualitative variables are represented as number (percentage), while quantitative variables are shown in mean (standard deviation). **Abbreviations:** PANSS: Positive and Negative Syndrome Scale; N/A: not available; SD: Standard deviation

A Quality assessments according to risk of bias**B As percentage (intention-to-treat)****Fig. 2** Quality assessments. **(A)** According to risk of bias for each study. **(B)** According to risk of bias as percentage**Results of the efficacy outcomes****Psychiatric outcomes**

Four studies, including 369 patients, reported the efficacy of sulforaphane on the psychiatric outcomes. There was no statistically significant difference between the sulforaphane and the placebo groups for the following outcomes: PANSS total score (MD= -1.55, 95% CI [-3.93, 0.82], P-value=0.2, $I^2=0\%$), PANSS Positive subscale (MD=0.76, 95% CI [-0.33, 1.85], P-value=0.17, $I^2=24\%$), PANSS negative subscale (MD= -1.42, 95% CI [-3.73, 0.89], P-value=0.23, $I^2=73\%$), PANSS Cognition (MD=0.09, 95% [-0.82, 1], P-value=0.85, $I^2=49\%$),

and PANSS depression (MD=0.18, 95% CI [-0.56, 0.92], P-value=0.63, $I^2=61\%$). Sulforaphane led to a statistically significant improvement in the General Psychopathology subscale (MD= -1.5, 95% CI [-2.78, -0.23], P-value=0.02), with no detectable heterogeneity ($P=0.16$, $I^2=0\%$). This analysis focused on the latest follow-up time points of the included studies, which ranged from 18 to 24 weeks, as shown in Fig. 3. Dose-based subgroup analyses for PANSS total and PANSS negative subscale were conducted according to sulforaphane dose. Sulforaphane showed a higher but not statistically significant trend regarding PANSS total score improvement at 136 $\mu\text{mol/}$

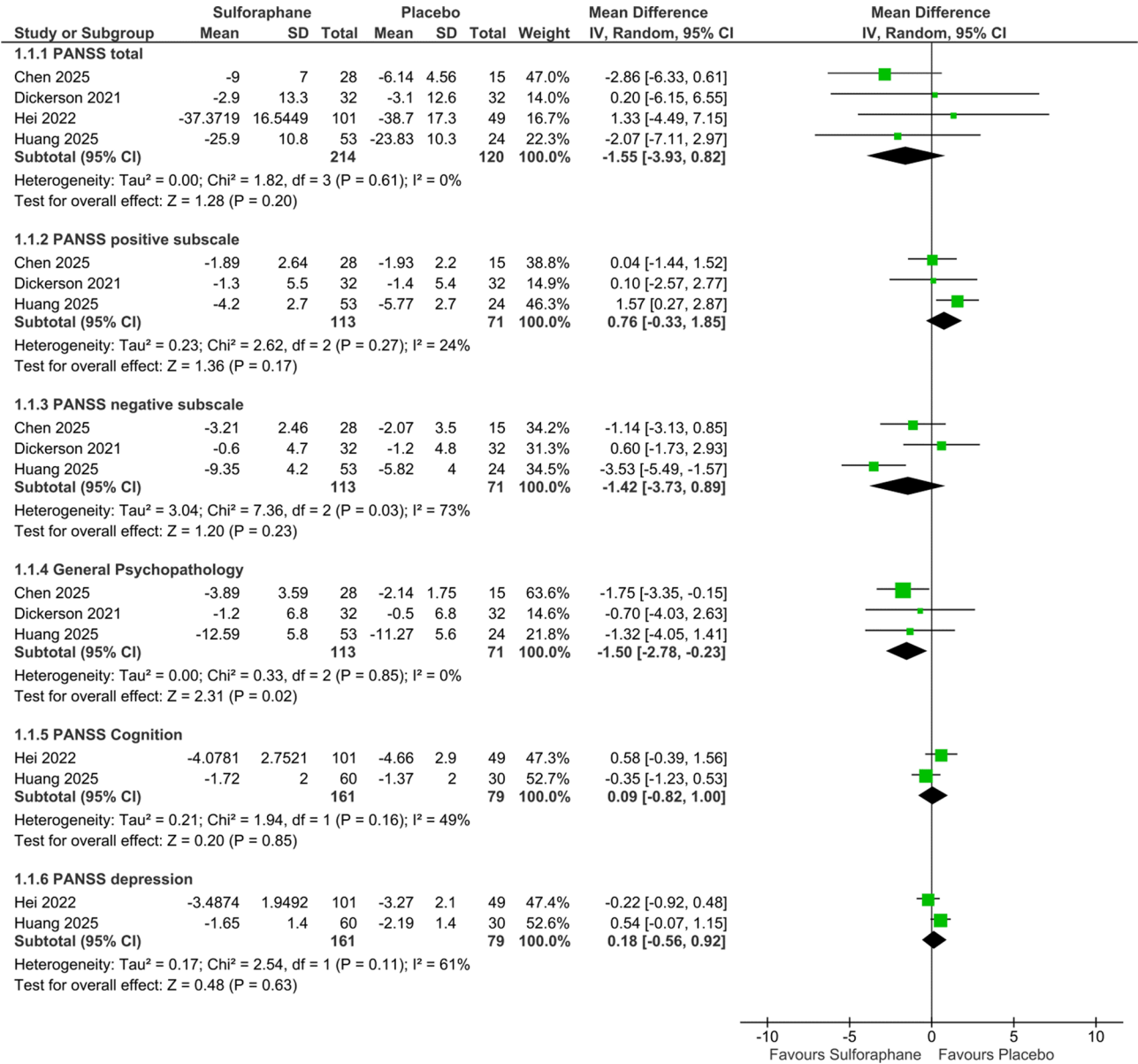


Fig. 3 Forest plot of the psychiatric outcomes at the latest follow-up point. (1) PANSS total, (2) PANSS Positive subscale, (3) PANSS Negative Subscale, (4) General Psychopathology, (5) PANSS Cognition, (6) PANSS depression. SD: Standard deviation; IV: Inverse-Variance, CI: Confidence interval

day (MD= -2.61, 95% CI [-5.47, 0.25], P-value=0.07, I²=0%), as shown in Supplementary Fig. 1. Regarding PANSS negative scale, sulfuraphane demonstrated a borderline beneficial effect at 136 μmol/day (MD= -2.34, 95% CI [-4.68, 0.00], P-value=0.05, I²=64%), as shown in Supplementary Fig. 1.

We conducted a sensitivity analysis by excluding one study at a time to investigate the source of heterogeneity in PANSS negative scale pooled effect. We found that heterogeneity was resolved by excluding Huang et al. 2025. The overall effect remained not significant (MD= -0.38, 95% CI [-2.07, 1.31], P-value=0.66, I²=19%), as shown in Supplementary Fig. 3. We conducted an analysis based

on a consistent 12-week point across all studies. Sulfuraphane significantly improved PANSS negative subscale (MD= -1.06, 95% CI= -1.95, -0.16], P-value=0.02), with no detectable heterogeneity (P=0.69, I²=0%). However, PANSS total score and PANSS Positive subscale changes were non-significant, as shown in Supplementary Fig. 4. Sulfuraphane showed a more significant improvement regarding PANSS negative subscale at the 12-week point analysis with 136 μmol/day dose (MD= -1.22, 95% CI [-2.18, -0.25], P-value=0.01, I²=0%), as shown in Supplementary Fig. 5.

Cognitive outcomes

Three studies reported the change of total MCCB score. Sulforaphane failed to show superiority over placebo regarding improving any of the pooled cognitive outcomes: MCCB total score (MD=0.35, 95% CI [-1.11, 1.82], P-value=0.64, I²=0%), Spatial Working Memory (MD=2.64, 95% CI [-1.9, 7.17], P-value=0.25, I²=38%), Verbal Learning (MD=1.56, 95% CI [-1.65, 4.78], P-value=0.34, I²=0%), and Reasoning-Problem Solving (MD=1.61, 95% CI [-1.2, 4.41], P-value=0.26, I²=0%), as shown in Supplementary Fig. 6.

Results of safety and metabolic outcomes

Metabolic outcomes

Three studies reported the metabolic outcomes. Sulforaphane led to a statistically significant reduction in the following metabolic markers: LDL (MD= -0.25, 95% CI [-0.42, -0.07], P-value=0.005, I²=0%), triglycerides (MD=

-0.25, 95% CI [-0.42, -0.08], P-value=0.003, I²=0%), and cholesterol (MD= -0.35, 95% CI [-0.57, -0.12], P-value=0.002, I²=0%). No significant differences were observed regarding HDL (MD= -0.07, 95% CI [-0.14, 0.01], P-value=0.07, I²=0%) or glucose levels (MD= -0.04, 95% CI [-0.24, 0.16], P-value=0.68, I²=0%). There was no detectable heterogeneity in the pooled analysis of all the markers (I²=0%), as shown in Fig. 4.

Discontinuation causes

All four included studies reported all causes of discontinuation. The discontinuation rate was significantly lower in the sulforaphane group compared to placebo group (RR=0.68, 95% CI [0.49, 0.95], P-value=0.02, I²=0%), with no detectable heterogeneity, as shown in Fig. 5.

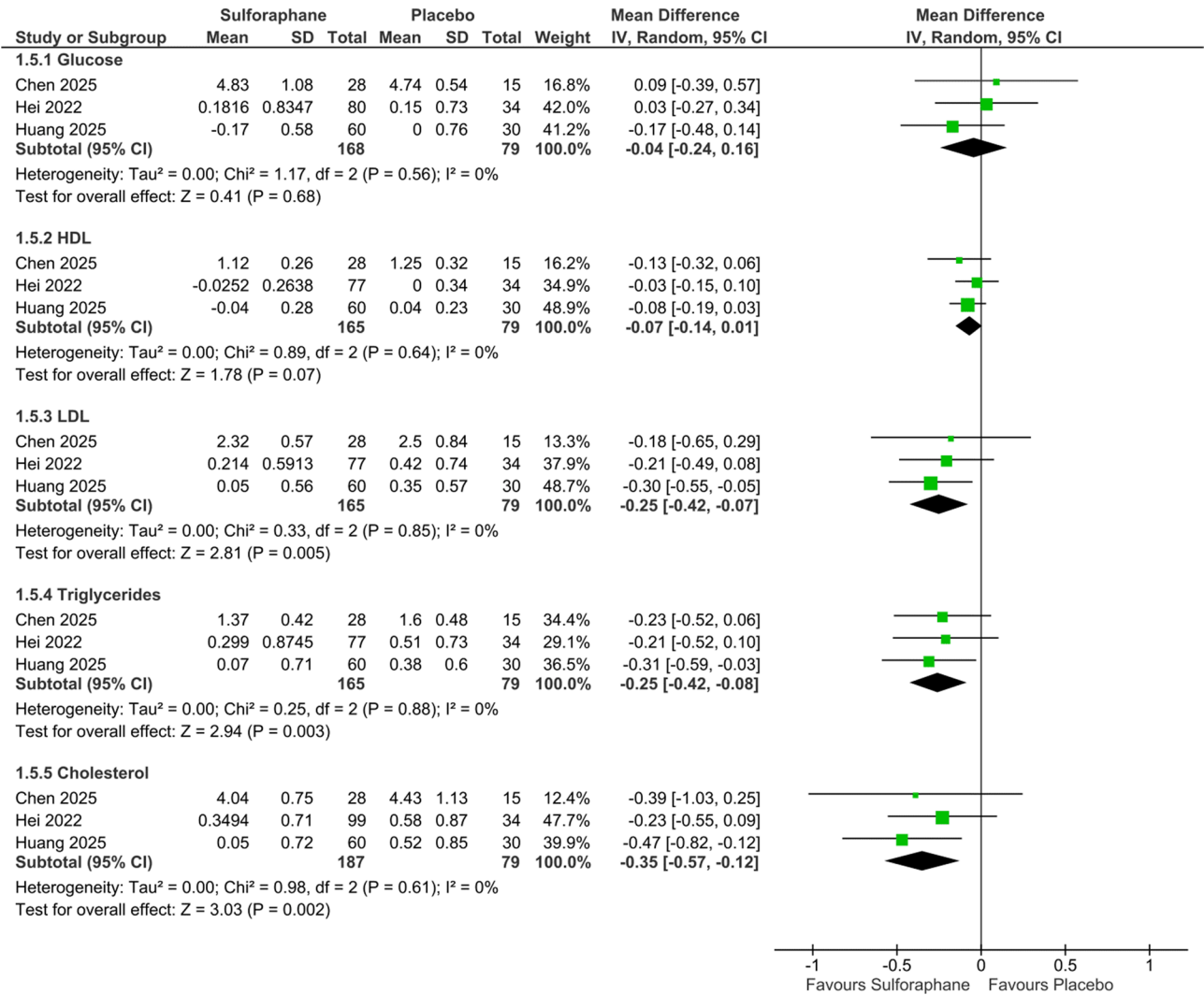


Fig. 4 Forest plot of the metabolic outcomes. (1) Glucose, (2) HDL, (3) LDL, (4) Triglycerides, (5) Cholesterol. IV: Inverse-variance; CI: Confidence interval

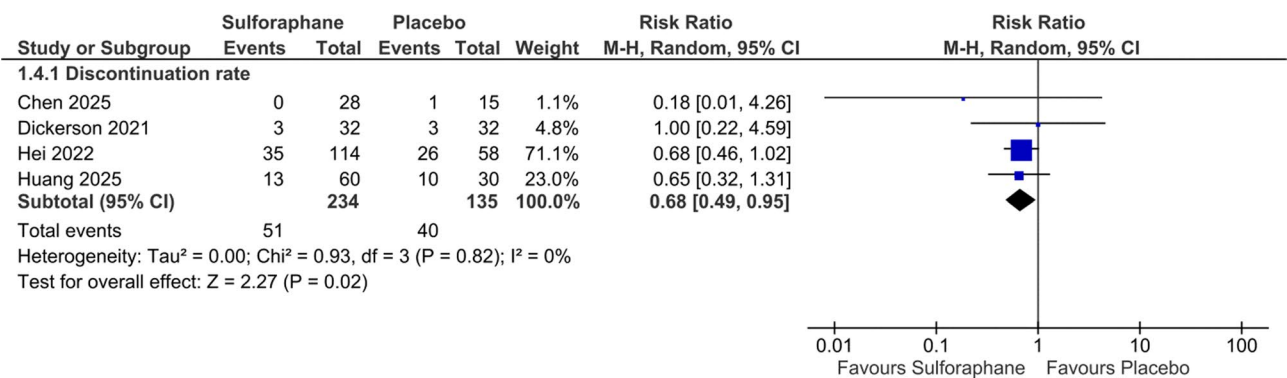


Fig. 5 Forest plot of all causes of discontinuation. M-H: Mantel-Hanzel; CI: Confidence interval

Adverse events

The studies included did not provide sufficient data regarding adverse events to be pooled in a meta-analysis. However, all included studies reported that sulforaphane was well-tolerated. In Dickerson et al., there were 2 serious adverse events in the sulforaphane group involving 2 patients, and 6 serious adverse events in the placebo group involving 5 participants. Hei et al. reported no significant differences between sulforaphane and placebo on the Abnormal Involuntary Movement Scale (AIMS), Barnes Akathisia Rating Scale (BARS), and Simpson–Angus Scale (SAS), and most items on the Systematic Assessment for Treatment Emergent Events (SAFTEE). Huang et al. found that sulforaphane was well tolerated and compared to placebo, as evaluated by the Treatment Emergent Symptom Scale (TESS). Similarly, Chen et al. reported no significant differences in TESS scores between sulforaphane and placebo.

Discussion

Our meta-analysis showed no significant difference between sulforaphane and placebo in improving PANSS total score and PANSS positive subscale with no significant heterogeneity in 12 weeks and 24 weeks. No individual study reported any meaningful improvement in either PANSS total or positive symptom scores. This lack of effect was consistent among all included studies despite variation in baseline PANSS total and positive scores and clinical presentation among the included studies. The baseline PANSS positive subscale ranged from 22 in the highest sample size to 12 in the lowest. This raises concerns about the potential of sulforaphane to exert any therapeutic benefit on positive symptoms.

The pooled analysis of the PANSS negative subscale in 12 weeks favored sulforaphane over placebo. However, the pooled analysis of the PANSS negative subscale in 24 weeks showed no significant difference. The analysis came with heterogeneity, which was resolved by leave-one-out sensitivity analysis when excluding Huang (2025). Huang et al. (2025) was the most significant study

regarding the reduction of the PANSS negative subscale. This may be explained by using a high dose of 136 μmol compared to other studies and came with a larger sample size than Chen (2025), which used the same dose. Also, the study was on patients who were diagnosed with schizophrenia for 10 years or less with prominent negative symptoms, and most of the participants had negative symptom scores higher than positive symptoms. On the contrary, Chen (2025) included patients who were diagnosed for 10 to 30 years, Hei (2022) included early-stage patients, and Dickerson (2021) was focused on schizophrenia or schizoaffective disorder patients with moderate psychotic symptoms. Dickerson (2021) was the study reporting the least improvement in PANSS negative symptoms score. The mean age of participants in Dickerson et al. ranged from 42 to 45 years while in Huang the mean age was around 23 years. Dickerson et al. study had higher baseline PANSS total scores and PANSS positive subscale than Huang et al. These variations demonstrate the need to identify the patients which may benefit the most from sulforaphane augmentation.

Dose-based subgroup analyses provided preliminary insights into a potential dose-response relationship. At the primary endpoint, sulforaphane at lower doses (66 and $\sim 100\ \mu\text{mol/day}$) did not show a statistically significant improvement in PANSS total score, but with the highest dose (136 $\mu\text{mol/day}$) a trend toward improvement was observed that did not reach statistical significance. Regarding the PANSS negative subscale, at the primary endpoint a borderline improvement ($P = 0.05$) was observed with the 136 $\mu\text{mol/day}$ dose, and a statistically significant improvement was observed at the 12-week analysis, while the 100 $\mu\text{mol/day}$ dose showed no effect. However, these observations should be interpreted with caution due to the limited number of trials and the small sample size. In an open-label study where 45 patients with schizophrenia received a high dose of sulforaphane 507.4 $\mu\text{mol/day}$, sulforaphane significantly improved the PANSS total and the PANSS negative subscale [33]. From a pharmacokinetic perspective,

sulforaphane is formed by the hydrolysis of its precursor glucoraphanin by the enzyme myrosinase or by gut microbiota [34]. It is metabolized via the mercapturic acid pathway, undergoing conjugation with glutathione and subsequent excretion as N-acetylcysteine derivatives in urine [35]. The absorption of sulforaphane is variable, ranging from ~ 1% to 40% among human volunteers [34]. Recent work by Zhu (2025) showed that combining glucoraphanin-rich extract with exogenous myrosinase and encapsulating it resulted in a conversion efficiency of ~ 72.1%, versus ~ 29.3% for unencapsulated delivery [36]. These pharmacological characteristics demonstrate that dose escalation alone may have limited benefit. Future studies should therefore not only evaluate higher doses but also consider optimized formulations to improve bioavailability while maintaining safety and tolerability.

The pooled analysis of general psychophysiology at the end of the study favored sulforaphane over placebo and came with no significant heterogeneity. Regarding cognition, even a previous study by Nouchi et al. (2021) reported a positive effect of sulforaphane on the cognition domain [37]. Our pooled effect of sulforaphane failed to exhibit significant improvements in cognitive outcomes, including both the overall MCCB total score and individual cognitive domains such as verbal learning, spatial working memory, and reasoning/problem-solving. However, Hei (2022) showed improvement in working memory, attention, and processing speed. This study came with the largest sample size and was in early-episode schizophrenia patients.

Although most individual studies included in our meta-analysis did not report significant improvements in metabolic markers, our pooled analysis showed that sulforaphane has a significant reduction in LDL, triglycerides, and cholesterol. The reduction can be attributed to the pathway of sulforaphane, which is a phytochemical bioactive substance [38]. It is reported as an activator of nuclear factor erythroid-2-related factor (Nrf2), which protects cells against oxidative damage by activation of Nrf2. Sulforaphane enhances mitochondrial function, as suggested by the decrease in ROS, the increase in mitochondrial membrane potential, and the upregulation of ATP production [39, 40], which contribute to ameliorating aberrant lipid metabolism and may contribute to reducing weight and controlling diabetes mellitus and hyperlipidemia, which are more prominent risks of some of antipsychotics [41, 42]. However, while the reductions were statistically significant, the magnitude of the change was modest. The limited clinical significance raises concerns about whether these changes may translate to cardiovascular benefits. Further studies are needed to determine the metabolic benefits of sulforaphane.

The discontinuation was less in the sulforaphane group compared to the placebo group. The discontinuation

causes were due to the withdrawal of consent, poor compliance with regular antipsychotics, and some changed to another medication. However, some studies did not clearly specify the reasons for participants dropout. All included studies reported that sulforaphane was well-tolerated, with no significant adverse effects.

Strength points and limitations

Our study is the first up-to-date meta-analysis to evaluate the efficacy of sulforaphane in schizophrenia patients, encompassing data from four RCTs. The studies were of high quality with no high risk of bias. The analysis was conducted at different time points, including the latest follow-up of the studies and a consistent 12-week point. The metabolic marker analysis showed no heterogeneity. Our study showed that the discontinuation was more in the placebo group than in the sulforaphane group, which indicates that the drug is well tolerated.

However, our study was not without limitations. The limited number of RCTs and small sample size of 369 patients is a significant limitation. Also, the schizophrenia patients included in the studies were not homogenous. There was a study with early-episode patients [22] and others with long-standing patients [20, 21]. The duration of the trials and doses was different, which could explain the heterogeneity found in some of the analyses. Also, the geographical concentration, as all studies were conducted in China except for Dickerson (2021) which was conducted in the USA. The heterogeneity and small sample size may limit the generalizability of our results. The adverse effects of sulforaphane are not well explained.

Significance of the findings and future research

Our study is the first to provide a contribution to the literature about the efficacy and safety of sulforaphane in schizophrenia. While the overall psychiatric and cognitive outcomes did not favor sulforaphane over the placebo. Sulforaphane was associated with statistically significant improvement in general psychopathology symptoms. The effect on negative symptoms is unclear. While there was no improvement at the latest follow-up analysis, there was a statistically significant modest improvement at 12 weeks with no heterogeneity.

Sulforaphane was also associated with favorable changes in the pooled metabolic markers, including reduction in LDL, TG, and cholesterol. These findings may be relevant if further explored, given the high cardiometabolic risks in patients with schizophrenia. The reduced discontinuation rate may indicate better overall tolerability.

We recommend further research be conducted with larger sample sizes with standardized treatment duration and consistent antipsychotic co-administration to confirm the long-term efficacy and safety of sulforaphane

in schizophrenia. Consider the specific patients who may benefit the most from sulforaphane as an adjunct in schizophrenia. Additionally, studies should explore the effect of sulforaphane on cognitive parameters and give clear data about the adverse effects and safety of the drug. Future trials in schizophrenia may consider a higher dose of the sulforaphane preparation $\geq 136 \mu\text{mol/day}$ and consider optimized formulations to enhance the bioavailability of sulforaphane.

Conclusion

In conclusion, our meta-analysis findings provide a valuable initial overview regarding the efficacy of sulforaphane in schizophrenia. While the overall findings did not demonstrate consistent improvements across various psychiatric outcomes, there was a modest effect on general psychopathology and negative symptoms at 12 weeks. Sulforaphane also appeared to produce favorable metabolic changes and a lower discontinuation rate. The favorable effects on LDL, triglycerides, and cholesterol are remarkable given the high cardiometabolic risks in this population. Given the limited number of studies and the heterogeneity. These results should be interpreted with caution. More well-powered, longer-term trials are needed to better understand sulforaphane's potential therapeutic role, particularly in relation to negative symptoms and metabolic health in schizophrenia. Future research should also focus on identifying specific patient subgroups that may benefit the most from sulforaphane augmentation.

Supplementary Information

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Supplementary Material 1

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Author contributions

Omar Kassar: Conceptualization, idea validation, supervision, screening, data extraction, risk of bias assessment, analysis, and writing—review and editing. Mohamed Ezzat M. Mansour: Writing—original draft, reviewing, risk of bias assessment, and validation. NourAllah Farag: Conceptualization, analysis, screening, and writing. Abdullah Selim: Data extraction, writing, and risk of bias assessment. Yasser Kewia: Methodology, screening, and draft writing. Obai Yousef: Data extraction, screening, writing—review and editing. Osama Hassan: Idea validation, writing—review and editing.

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Data availability

The data that support the findings of this study are available from the corresponding author upon request.

Declarations

Ethics approval and consent to participate

This article does not contain any studies with human participants or animals performed by any of the authors.

Consent to participate

Not applicable.

Consent for publication

All authors' consent has been obtained by the corresponding author for publication.

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Competing interests

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References

1. Velligan DI, Rao S. The epidemiology and global burden of schizophrenia. *J Clin Psychiatry*. 2023 Jan;84(1) <https://doi.org/10.4088/JCPMS21078COM5>
2. Hany M, Rizvi A. Schizophrenia. StatPearls. Treasure Island (FL): StatPearls Publishing; 2025.
3. Patel KR, Cherian J, Gohil K, Atkinson D. Schizophrenia: overview and treatment options. *PT*. 2014 Sep;39(9):638–645.
4. Gozdzik-Zelazny A, Borecki L, Pokorski M. Depressive symptoms in schizophrenic patients. *Eur J Med Res*. 2011 Dec;16(12):549–52. <https://doi.org/10.186/2047-783x-16-12-549>
5. McCutcheon RA, Keefe RSE, McGuire PK. Cognitive impairment in schizophrenia: aetiology, pathophysiology, and treatment. *Mol Psychiatry*. 2023 May;28(5):1902–18. <https://doi.org/10.1038/s41380-023-01949-9>
6. Bai W, et al. Worldwide prevalence of suicidal ideation and suicide plan among people with schizophrenia: a meta-analysis and systematic review of epidemiological surveys. *Transl Psychiatry*. 2021 Oct;11(1):552. <https://doi.org/10.1038/s41398-021-01671-6>
7. Tanskanen A, Tiihonen J, Taipale H. Mortality in schizophrenia: 30-year nationwide follow-up study. *Acta Psychiatr Scand*. 2018 Dec;138(6):492–499. <https://doi.org/10.1111/acps.12913>
8. Keepers GA, et al. The American Psychiatric Association practice guideline for the treatment of patients with schizophrenia. *Am J Psychiatry*. 2020 Sep;177(9):868–872. <https://doi.org/10.1176/appi.ajp.2020.177901>
9. Buchanan RW, et al. The Cognitive and Negative Symptoms in Schizophrenia Trial (CONSIST): the efficacy of glutamatergic agents for negative symptoms and cognitive impairments. *Am J Psychiatry*. 2007 Oct;164(10):1593–1602. <https://doi.org/10.1176/appi.ajp.2007.06081358>
10. Kassar O, et al. Efficacy and safety of ondansetron in schizophrenia: a systematic review and meta-analysis of randomized controlled trials. *Gen Hosp Psychiatry*. 2025 Jun;96:37–46. <https://doi.org/10.1016/j.genhosppsych.2025.06.001>
11. Stogios N, et al. Autonomic nervous system dysfunction in schizophrenia: impact on cognitive and metabolic health. *NPJ Schizophr*. 2021 Apr;7(1):22. <https://doi.org/10.1038/s41537-021-00151-6>
12. Prieto SG, Almeida MC, Silva JCS, Del-Bel E, Echeverry MB. Extrapyramidal side effects with chronic atypical antipsychotic can be predicted by labeling pattern of FosB and phospho Thr34-DARPP-32 in nucleus accumbens. *Biomedicines*. 2023 Sep;11(10):1429. <https://doi.org/10.3390/biomedicines11102677>
13. Pillinger T et al. Comparative effects of 18 antipsychotics on metabolic function in patients with schizophrenia, predictors of metabolic dysregulation, and association with psychopathology: a systematic review and network meta-analysis. *Lancet Psychiatry*. 2020 Jan;7(1):64–77. [https://doi.org/10.1016/S2215-0366\(19\)30416-X](https://doi.org/10.1016/S2215-0366(19)30416-X)
14. Vanduchova A, Anzenbacher P, Anzenbacherova E. Isothiocyanate from broccoli, sulforaphane, and its properties. *J Med Food*. 2019 Feb;22(2):121–6. <https://doi.org/10.1089/jmf.2018.0024>

15. Carletti B, Banaj N, Piras F, Bossù P. Schizophrenia and glutathione: a challenging story. *J Pers Med*. 2023 Oct;13(11):1717. <https://doi.org/10.3390/jpm13111526>
16. Hung C-N, Huang H-P, Wang C-J, Liu K-L, Lii C-K. Sulforaphane inhibits TNF- α -induced adhesion molecule expression through the Rho A/ROCK/NF- κ B signaling pathway. *J Med Food*. 2014 Oct;17(10):1095–102. <https://doi.org/10.1089/jmf.2013.2901>
17. Folbergrová J, Ješina P, Otáhal J. Protective effect of sulforaphane on oxidative stress and mitochondrial dysfunction associated with status epilepticus in immature rats. *Mol Neurobiol*. 2023 Apr;60(4):2024–35. <https://doi.org/10.1007/s12035-022-03201-x>
18. Shirai Y, Fujita Y, Hashimoto K. Effects of the antioxidant sulforaphane on hyperlocomotion and prepulse inhibition deficits in mice after phencyclidine administration. *Clin Psychopharmacol Neurosci*. 2012 Aug;10(2):94–8. <https://doi.org/10.9758/cpn.2012.10.2.94>
19. Shiina A, et al. An open study of sulforaphane-rich broccoli sprout extract in patients with schizophrenia. *Clin Psychopharmacol Neurosci*. 2015 Apr;13(1):62–7. <https://doi.org/10.9758/cpn.2015.13.1.62>
20. Chen H, et al. Effects of sulforaphane on negative symptoms and cognitive impairments in chronic schizophrenia patients: a randomized double-blind trial. *J Psychiatr Res*. 2025 Apr;184:464–72. <https://doi.org/10.1016/j.jpsychires.2025.03.023>
21. Huang J, et al. Efficacy and safety of sulforaphane added to antipsychotics for the treatment of negative symptoms of schizophrenia: a randomized controlled trial. *J Clin Psychiatry*. 2025 Jan;86(1). <https://doi.org/10.4088/JCP.24m15272>
22. Hei G, et al. Sulforaphane effects on cognition and symptoms in first and early episode schizophrenia: a randomized double-blind trial. *Schizophr Bull Open*. 2022 Jan;3(1):sgac024. <https://doi.org/10.1093/schizbullopen/sgac024>
23. Dickerson F, et al. Randomized controlled trial of an adjunctive sulforaphane nutraceutical in schizophrenia. *Schizophr Res*. 2021 May;231:142–4. <https://doi.org/10.1016/j.schres.2021.03.018>
24. Guo J et al. Effects of sulforaphane on ABC and SRS scales in patients with autism spectrum disorder: a meta-analysis. *Brain Dev*. 2025 Apr;47(2):104321. <https://doi.org/10.1016/j.braindev.2025.104321>
25. Fornaro M, et al. Nutraceuticals and phytochemicals in the treatment of schizophrenia: a systematic review and network meta-analysis 'Nutra NMA SCZ'. *Mol Psychiatry*. 2025 Jan;30(1):168–87. <https://doi.org/10.1038/s41380-024-02645-y>
26. Page MJ, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. *Int J Surg*. 2021 Apr;88:105906. <https://doi.org/10.1016/j.ijsu.2021.105906>
27. Cumpston M, et al., editors. Updated guidance for trusted systematic reviews: a new edition of the Cochrane Handbook for Systematic Reviews of Interventions. *Cochrane Database Syst Rev*. 2019 Oct;10(10):ED000142. <https://doi.org/10.1002/14651858.ED000142>
28. Ouzzani M, Hammady H, Fedorowicz Z, Elmagarmid A. Rayyan—a web and mobile app for systematic reviews. *Syst Rev*. 2016 Dec;5(1):210. <https://doi.org/10.1186/s13643-016-0384-4>
29. Higgins JPT, et al. The Cochrane Collaboration's tool for assessing risk of bias in randomised trials. *BMJ*. 2011 Oct;343:d5928. <https://doi.org/10.1136/bmj.d5928>
30. Yang Z, Lim K, Lam M, Keefe R, Lee J. Factor structure of the positive and negative syndrome scale (PANSS) in people at ultra high risk (UHR) for psychosis. *Schizophr Res*. 2018 Nov;201:85–90. <https://doi.org/10.1016/j.schres.2018.05.024>
31. August SM, Kiwanuka JN, McMahon RP, Gold JM. The MATRICS consensus cognitive battery (MCCB): clinical and cognitive correlates. *Schizophr Res*. 2012 Jan;134(1):76–82. <https://doi.org/10.1016/j.schres.2011.10.015>
32. Egger M, Smith G, Schneider M, Minder C. Bias in meta-analysis detected by a simple, graphical test. *BMJ*. 1997 Sep;315(7109):629–34. <https://doi.org/10.1136/bmj.315.7109.629>
33. Zeng J, et al. Zeng J, et al. The association of SOD and HsCRP with the efficacy of Sulforaphane in schizophrenia patients with residual negative symptoms. *Eur Arch Psychiatry Clin Neurosci*. Aug. 2024;274(5):1083–92. <https://doi.org/10.1007/s00406-023-01679-7>
34. Fahey JW, Holtzclaw WD, Wehage SL, Wade KL, Stephenson KK, Talalay P. Sulforaphane bioavailability from glucoraphanin-rich broccoli: control by active endogenous myrosinase. *PLoS One*. 2015 Nov;10(11):e0140963. <https://doi.org/10.1371/journal.pone.0140963>
35. Zhang Y, Tang L. Discovery and development of sulforaphane as a cancer chemopreventive phytochemical. *Acta Pharmacol Sin*. 2007 Sep;28(9):1343–54. <https://doi.org/10.1111/j.1745-7254.2007.00679.x>
36. Zhu W, Cremonini E, Mastaloudis AF, Mitchell AE, Bornhorst GM, Oteiza PI. Optimization of sulforaphane bioavailability from a glucoraphanin-rich broccoli seed extract in a model of dynamic gastric digestion and absorption by Caco-2 cell monolayers. *Food Funct*. 2025 Jan;16(1):314–28. <https://doi.org/10.1039/d4fo04561k>
37. Nouchi R, Hu Q, Saito T, Kawata NYDS, Nouchi H, Kawashima R. Brain training and sulforaphane intake interventions separately improve cognitive performance in healthy older adults, whereas a combination of these interventions does not have more beneficial effects: evidence from a randomized controlled trial. *Nutrients*. 2021 Jan;13(2):404. <https://doi.org/10.3390/nu13020352>
38. Kiser C, Gonul CP, Olcum M, Genc S. Inhibitory effects of sulforaphane on NLRP3 inflammasome activation. *Mol Immunol*. 2021 Dec;140:175–85. <https://doi.org/10.1016/j.molimm.2021.10.014>
39. Lei P, et al. Sulforaphane improves lipid metabolism by enhancing mitochondrial function and biogenesis in vivo and in vitro. *Mol Nutr Food Res*. 2019 Feb;63(4):e1800795. <https://doi.org/10.1002/mnfr.201800795>
40. Zhao L, et al. Protective role of sulforaphane in lipid metabolism-related diseases. *Mol Biol Rep*. 2025 Feb;52(1):241. <https://doi.org/10.1007/s11033-025-10358-w>
41. Cohen D. Atypical antipsychotics and new onset diabetes mellitus. An overview of the literature. *Pharmacopsychiatry*. 2004 Jan;37(1):1–11. <https://doi.org/10.1055/s-2004-815468>
42. Meyer JM, Koro CE. The effects of antipsychotic therapy on serum lipids: a comprehensive review. *Schizophr Res*. 2004 Sep;70(1):1–17. <https://doi.org/10.1016/j.schres.2004.01.014>

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